EliteDesign: Elite-Guided Learning for RNA Design via the Cross-Entropy Method

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Abstract

The aim of RNA design is to find, within an exponentially large search space, a nucleotide sequence that folds into a given target structure. Efficient RNA design could aid in the development of RNA-based technologies such as riboswitches and ribozymes. However, finding sequences that fold into the target structure with high probability remains challenging, particularly for long RNAs. To address this challenge, we introduce an approach that uses the cross-entropy method to iteratively refine a distribution over valid RNA sequences. Our method outperforms state-of-the-art RNA design methods in average target folding probability, achieving over 0.60 on Eterna100, with a runtime of seven minutes for the entire benchmark per random seed. It also scales to longer RNA structures, reaching an average folding probability of 0.51 on 10 16S rRNA targets ranging from 950 to 1525 nucleotides, compared with 0.20 for SAMFEO, while achieving a lower average runtime per puzzle.